VentureTruth: Automated Verification of Startup Pitch Claims

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1. Context: Startup pitch decks are high-volume, self-promotional documents

- Venture Capital investors on average screen **thousands** of pitch decks per year
- These pitch decks contain claims that mix facts, internal metrics, and promotional language.
- Screening these documents can be inefficient and requires analysts to do verification of startup claims manually

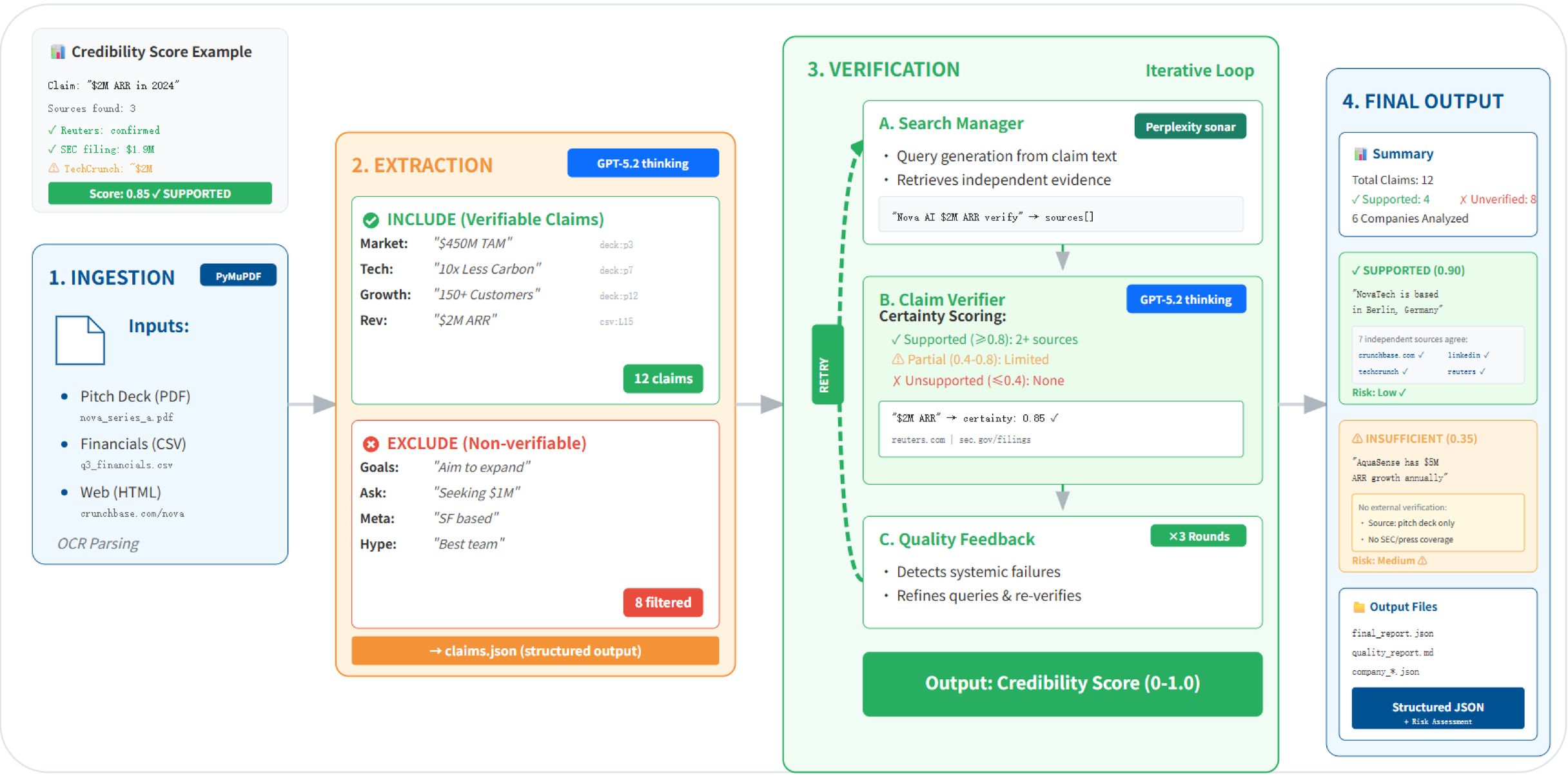
→ *Persistent information asymmetry between founders and investors.*

2. Research Problem: How effectively can LLMs verify startup claims?

Core Approach The contributions of this work include:

- The contributions of this work include: We evaluate the ability of large language models to extract and verify structured metadata from heterogeneous startup artifacts.
- We study the effectiveness of large language models for identifying, extracting, and verifying claims, with a focus on multi-source, iterative verification.
- We propose a novel end-to-end architecture for processing startup pitch decks that integrates claim extraction, evidence retrieval, iterative verification, and robustness auditing.

3. Solution: LLM-Based Claim Verification Pipeline for VC Due Diligence



An end-to-end NLP pipeline for extracting and verifying factual claims from startup pitch decks.

Key Innovations

- **Novel application of Large Language Models (LLMs)** to systematically verify self-promotional claims in startup pitch materials.
- **Evaluation of LLMs as claim verification tools**, assessing classification into *Supported*, *Contradicted*, or *Insufficient Evidence*.
- **Verification with explanations**: Natural language explanations support for calibrating investor decision-making, such as downstream screening.

4. Evaluation of Results

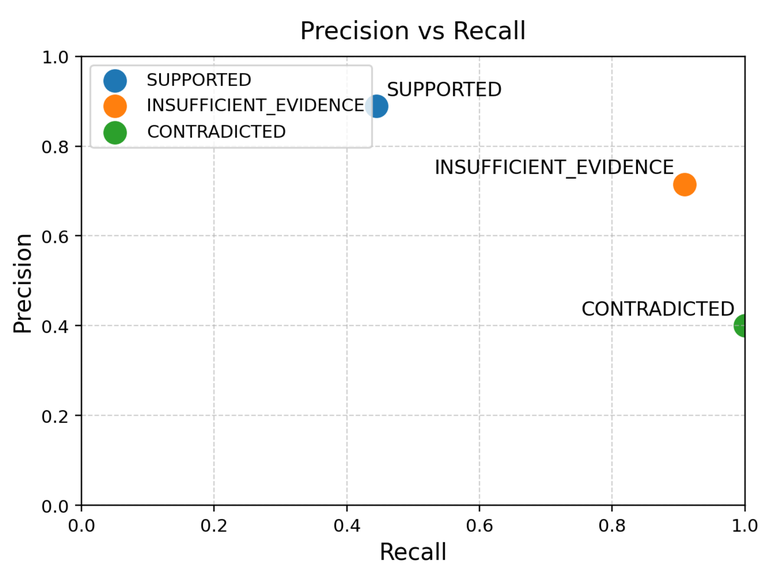


Figure 1: The Supported class achieves high precision, whereas the Contradicted class prioritizes recall.

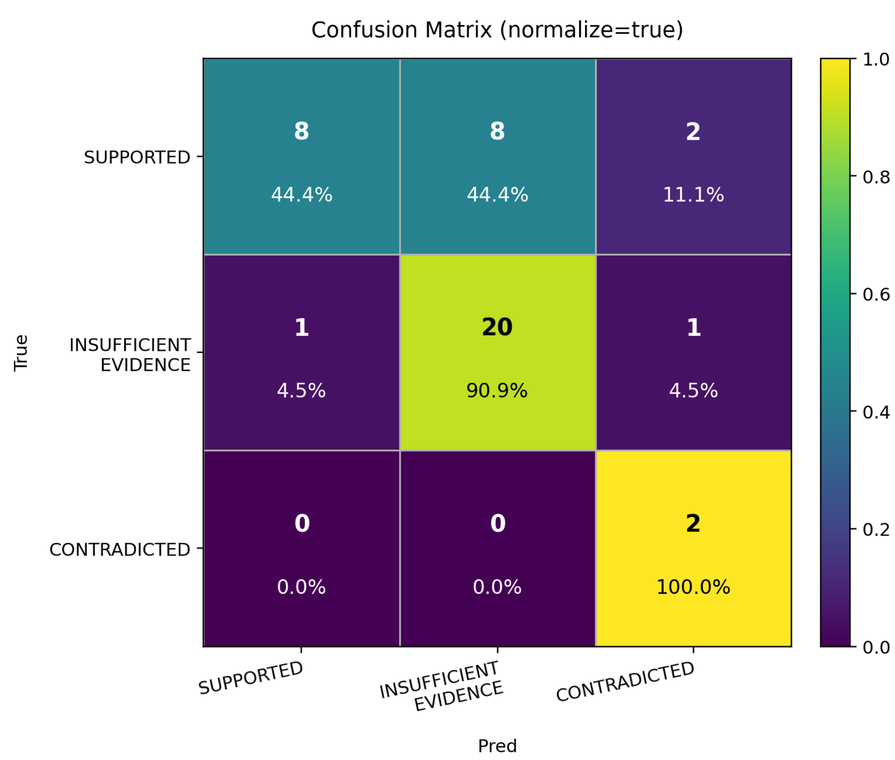


Figure 2: High accuracy is observed for Insufficient Evidence (90.9%) and Contradicted (100%) classes.

Github

5. Impact & Future Directions

Why It Matters

For Investors

DD time: Weeks → Hours

For Research

Cornerstone LLM-Based Startup Claim Verification

Future Work Directions:

1. **Workflow Customization:** Allow model selection for each individual step.
2. **Dataset Expansion:** Increase sample sizes for more robust evaluation.
3. **Broader LLM Benchmarking:** Test additional models, including open-source LLMs.
4. **Multimodal Integration:** Extend extraction to video and audio formats.

LaTeX TikZposter